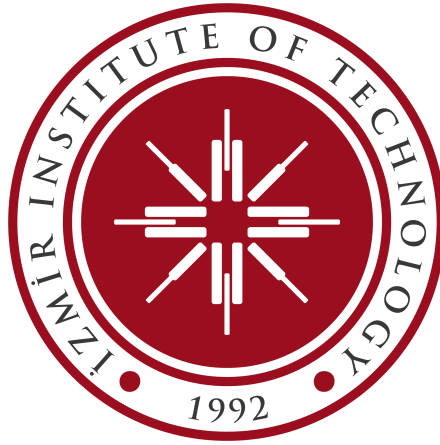


CENG424 Embedded Computer Systems
Project Proposal
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Smart Medication Sorting and Monitoring System

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Abstract

This project aims to design and implement an embedded system for the automated sorting and monitored dispensing of medications. While existing solutions primarily offer digital reminders, they fail to address the physical handling of mixed medications. The proposed system bridges this gap by utilizing a microcontroller-based architecture to physically classify, route, and dispense pills. Medications loaded in bulk are sequentially analyzed by a TCS3200 color sensor. An Arduino Uno processes these RGB values and actuates SG90 micro servo motors to physically rotate a sorting funnel, directing each pill into its designated internal compartment. For the dispensing phase, a DS3231 Real-Time Clock (RTC) strictly monitors the user-configured schedule via a 16x2 I2C LCD interface. At the prescribed time, a secondary gating mechanism releases the correct medication into an accessible tray, accompanied by an acoustic buzzer alert. Using colored candies as safe physical proxies, the prototype will demonstrate a fully closed-loop embedded system, validating real-time sensor processing, mechanical routing, and synchronized scheduling. Overall, the system reduces human error, improves medication adherence, and enhances patient safety through automated sorting and timed dispensing.

1 Introduction

This project presents a microcontroller-based embedded system designed for the automated classification, secure storage, and scheduled dispensing of medications. Managing multiple medications at different times of the day is a challenging task, and manual organization may lead to incorrect dosing or missed schedules. The proposed system aims to reduce such risks by integrating physical sensing, timing, and electromechanical actuation within a single automated pipeline. The core novelty of this system lies in how it physically handles the items. Operationally, the workflow is divided into two primary phases: classification and dispensing.

During the classification phase, bulk-loaded items fall sequentially into a sensing chamber where a color sensor reads their physical profiles. Based on a predefined logic map, the microcontroller triggers a primary routing mechanism, a 3D-printed chute driven by a servo motor, which physically rotates to align with the appropriate storage receptacle. Once aligned, the item is pushed into the compartment.

In the dispensing phase, an integrated Real-Time Clock (RTC) continuously monitors the schedule. Upon reaching a target timestamp, a secondary gating servo opens the specific compartment's base, dropping the designated item into a user-accessible tray while triggering an acoustic notification. By automating both the physical sorting mechanisms and the temporal tracking, the proposed system demonstrates a practical, end-to-end implementation of real-time embedded control.

2 Problem Definition

Managing a complex daily medication regimen is a significant challenge, particularly for elderly patients or individuals with chronic conditions who rely on manual sorting. Distributing various pills into traditional weekly organizers is a tedious process that is highly susceptible to human error, especially when different medications share similar shapes and differ only slightly in color.

The consequences of such mistakes, including taking the wrong medication, missing a critical dose, or accidentally double-dosing, can lead to serious health complications. While existing technological solutions, such as smartphone alarms or smart pillboxes with LED indicators, remind users *when* to take their medication, they entirely neglect the physical organization aspect. The user is still burdened with identifying and manually pre-sorting the correct medications from a mixed set, leaving a critical gap where dangerous errors can occur.

The fundamental problem addressed in this project is the lack of a unified system that eliminates the manual pre-sorting phase. There is a critical need for an embedded system capable of taking a mixed batch of medications, classifying them autonomously based on measurable physical properties, isolating them, and then dispensing them precisely on time. Our proposed system introduces a novel approach by removing the user from the physical sorting equation, thereby providing a significantly safer and more reliable medication management solution.

3 Literature Review

Medication non-adherence and medication handling errors have motivated the development of smart medication adherence systems that combine storage, reminders, and monitoring capabilities. Faisal et al. (2021) define smart medication adherence products as systems that incorporate connectivity and automated data processing. Their review shows that most existing solutions focus on reminder mechanisms, alarms, and user notifications rather than physical medication handling. An updated scoping review by the same research group Faisal et al. (2023) further confirms that the field has expanded rapidly, but most designs still assume that medications are manually pre-sorted before use.

Several embedded systems have been proposed to support medication management through guidance and reminders. For example, Bindu Sree et al. (2020) introduce a smart pill box that integrates audio prompts, display output, and LED indicators to help users select the correct compartment at the correct time. This approach improves usability and accessibility, but it still relies on pre-arranged compartments and does not solve the problem of physically sorting mixed items.

More advanced systems focus on monitoring and verification. Jimenez et al. (2024) present a medication dispenser with a touch screen and a consumption authenticator to support adherence monitoring in elderly users. This improves reliability by checking whether medication use actually occurred, but the system still addresses the intake phase more than the organization phase.

In addition to medication-focused systems, sensor-based object sorting approaches have been explored in embedded environments. Sinha et al. (2020) present a color sensor-based robotic arm that classifies objects using real-time sensor data and directs them to designated locations. While such systems demonstrate effective physical classification and actuation, they are not designed specifically for medication workflows and do not include scheduling or user interaction components required in healthcare applications.

Overall, existing studies demonstrate significant progress in scheduling, reminders, monitoring, and object classification, but the physical sorting of mixed medications is not explicitly addressed in most existing systems. Most existing systems assume that the pills are already organized correctly before they are loaded into the device.

In contrast, the proposed Smart Medication Sorting and Monitoring System addresses this gap by combining physical sorting with time-based dispensing in a single embedded platform. By using sensor-based classification and sequential processing, the system reduces dependence on manual pre-sorting and offers a practical approach that bridges medication organization and adherence support.

4 Stages

The development of the proposed system will be executed in six sequential stages to ensure a systematic implementation of both hardware and software components:

1. The physical structure of the prototype will be designed and assembled. This includes developing a hopper mechanism for bulk feeding, internal storage compartments, and a final dispensing tray to ensure controlled and sequential processing of items.
2. A color sensor module will be integrated to measure the physical properties of incoming items. Calibration will be performed to ensure reliable distinction between different categories under stable, enclosed lighting conditions.
3. A decision-making mechanism and a User Interface (UI) will be implemented. The UI will feature a 16x2 I2C LCD screen to display system status (e.g., "Sorting," "Idle," "Dispense Time") and tactile push buttons for users to easily configure daily medication schedules and acknowledge alerts.
4. Microcontroller-driven servo motors will be mechanically linked and programmed to route classified items into designated storage compartments based on real-time decisions derived from the sensor data.
5. An RTC module will be incorporated to track predefined schedules. In addition, a finite state machine (FSM) based control logic will be developed to manage system states. An acoustic buzzer and a secondary actuation mechanism will be integrated to release medications into the user tray exactly at the configured times.
6. The fully integrated system will be tested and validated. Key performance metrics such as sorting accuracy, mechanical reliability, and timing precision will be evaluated through repeated experimental trials.

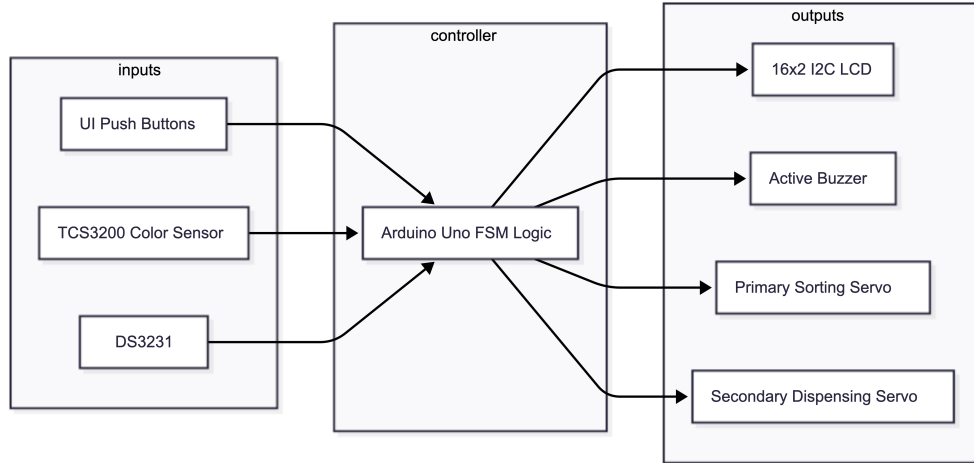


Figure 1: Basic System Diagram. The figure shows the system architecture. The inputs are user push buttons a color sensor and an RTC module. They send data to the Arduino Uno controller. The controller processes data and drives the outputs. The outputs are the LCD screen the active buzzer and the servo motors.

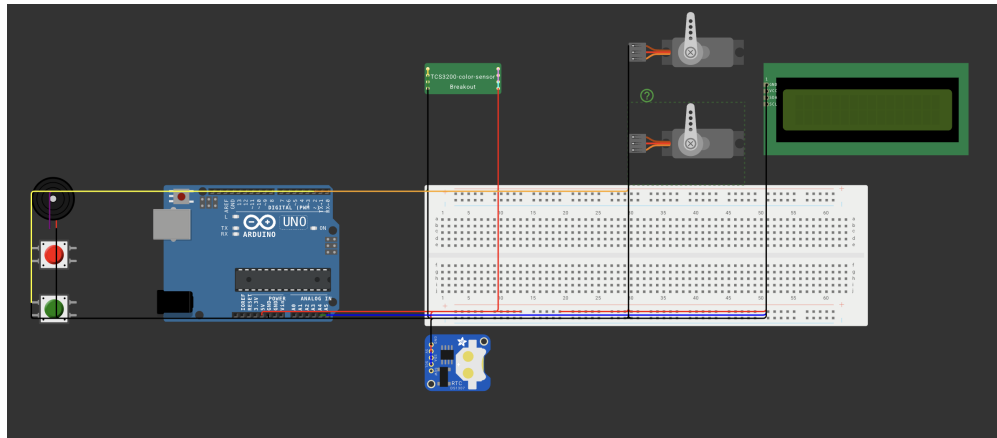


Figure 2: Basic Components. This figure shows the physical wiring. The Arduino Uno is the main central unit. It connects to the LCD screen and the RTC module using I2C pins. The servo motors and the buzzer connect to the digital pins of the Arduino. The push button connects to an input pin. All parts share the same power and ground connections.

5 Tools, Software, Hardware

The implementation of the proposed embedded system requires a set of hardware components, software environments, and mechanical design tools to ensure reliable sensing, measurement, and actuation.

- **Hardware Specifications:**

- *Microcontroller:* An Arduino Uno (ATmega328P) will serve as the central processing unit to process sensor data and execute control algorithms.
- *Sensors & Timing:* A TCS3200 color sensor will be utilized for RGB physical property mapping. A

DS3231 Real-Time Clock (RTC) module will provide high-precision, temperature-compensated time tracking via the I2C protocol.

- *Actuators & Feedback:* SG90 Micro servo motors controlled via PWM signals will actuate the primary sorting chute and secondary dispensing gates. An active 5V buzzer will provide acoustic feedback, and a 16x2 LCD display with an I2C backpack will serve as the visual output.

- **Software Integration:**

- *Firmware:* Developed in Embedded C/C++ using the Arduino IDE.
- *Architecture:* A non-blocking control structure based on a finite state machine (FSM) using `millis()` will be implemented to enable real-time multitasking without interrupting sensor readings or UI updates.
- *Libraries:* Standard libraries such as `Wire.h` (I2C communication), `Servo.h` (motor control), and `LiquidCrystal_I2C.h` (display management) will be integrated.

- **Mechanical Tools:**

- *Design:* Mechanical components (hopper, sorting channels, dispensing tray) will be designed using CAD tools such as Fusion 360 or Tinkercad.
- *Fabrication:* The prototype will be constructed using 3D-printed PLA parts for moving components to ensure structural stability, combined with acrylic or accessible plastics for the main chassis housing.

6 Experiments&Results, Test Phases

To comprehensively evaluate the performance, accuracy, and reliability of the proposed embedded system, a series of controlled experiments will be conducted. Due to safety and practical constraints, differently colored candies of uniform size and shape will be utilized as physical proxies for real medications. The experimental setup is divided into two main phases with strict configurations:

1. Classification and Sorting Accuracy Test

- **Objective:** To evaluate the precision of the TCS3200 color sensor, the accuracy of the RGB threshold mapping, and the mechanical reliability of the servo routing chute.
- **Setup:** A standardized batch of 90 items (30 pieces each of 3 distinct colors) will be bulk-fed into the system. To evaluate robustness, tests will be repeated under three varying ambient lighting conditions: bright room, dim room, and dark room.
- **Metrics:**
 - *Sorting Accuracy (True Positive Rate):* Percentage of correctly classified and routed items (Target: > 95%).
 - *Misclassification Rate:* Percentage of items routed to the wrong compartment.
 - *Mechanical Jam Rate:* Percentage of items that fail to dispense due to physical blockages in the hopper or chute (Target: < 2%).

2. Scheduling and Dispensing Reliability Test

- **Objective:** To evaluate the reliability of the FSM control logic, the drift precision of the DS3231 RTC module, and the consistency of the secondary dispensing mechanism.
- **Setup:** The system will undergo a 48-hour continuous operation test. Six distinct mock medication schedules will be configured using the UI buttons and LCD. The system will be left to operate autonomously.
- **Metrics:**
 - *Dispensing Success Rate:* Verification that the correct compartment's gating servo actuates without releasing adjacent items (Target: 100%).
 - *Temporal Precision:* The time drift between the configured schedule and the actual buzzer/dispensing event (Acceptable tolerance: ± 2 seconds).

7 Weekly Schedule

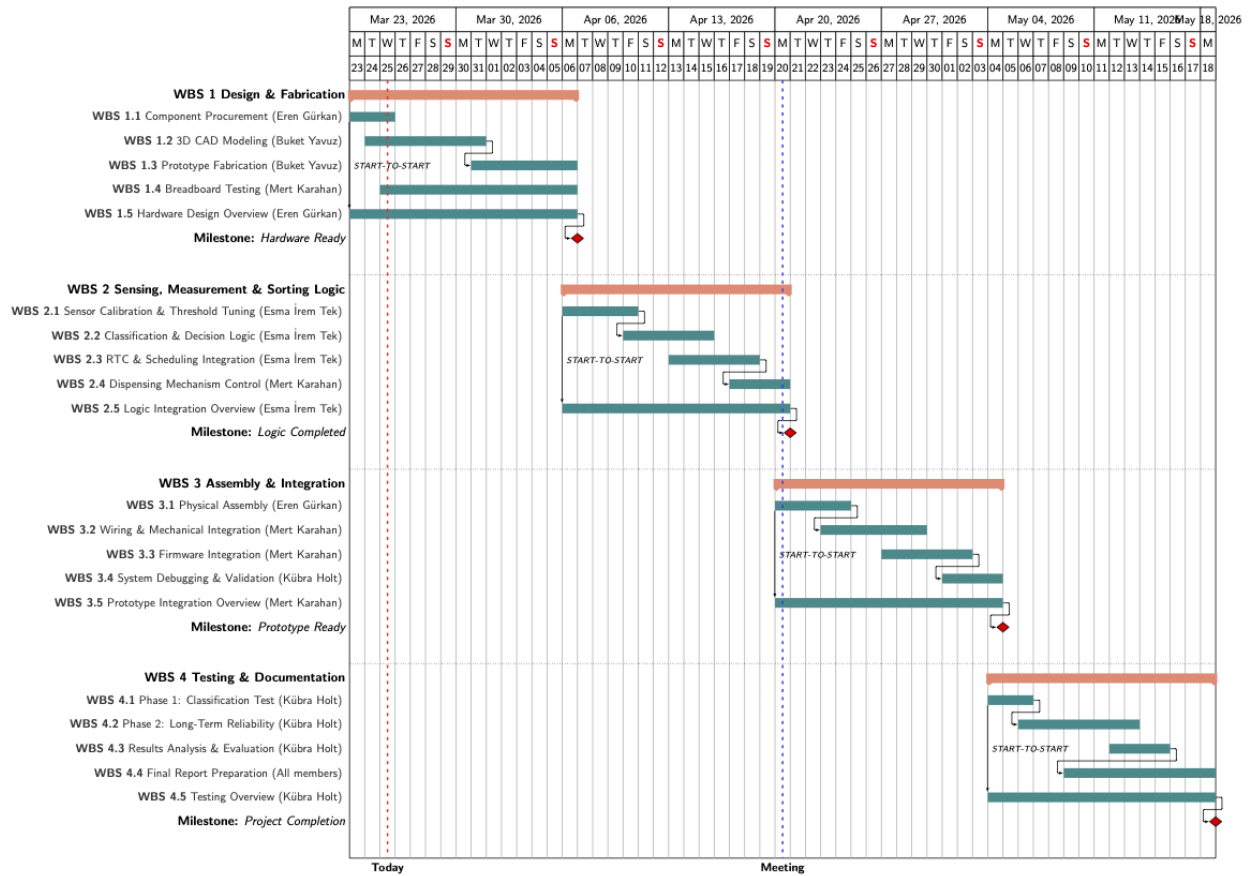


Figure 3: Smart Medication Sorting and Monitoring System: Weekly Implementation Schedule

Task Ownership & Assignments:

- **WBS 1 (Design & Fabrication):** Buket Yavuz & Eren Gürkan & Mert Karahan
- **WBS 2 (Sensing & Sorting Logic):** Esma İrem Tek & Mert Karahan
- **WBS 3 (Assembly & Integration):** Mert Karahan & Kübra Holt & Eren Gürkan
- **WBS 4 (Testing & Documentation):** Kübra Holt & All members

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